

## Final Group Assignment: AI-Agent Research Pipeline for Sustainable Portfolio Construction

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### Assessment format

This is a **group assignment**. Each group will design, prototype and defend an **AI-assisted investment research pipeline** that constructs a sustainable investment portfolio.

The assignment has two components:

| Component                                                  | Weight     |
|------------------------------------------------------------|------------|
| Written report / essay                                     | <b>60%</b> |
| Group presentation, pipeline demonstration and Q&A defence | <b>40%</b> |

The final presentation will take place during the scheduled exam session on **Friday 22 May 2026, 18:15–20:15**.

## 1. Assignment title

### Building a Sustainable Portfolio with an AI-Agent Research Pipeline

Note: This is an academic prototype and must not be presented as a regulated investment product, financial advice or a compliant Article 8 / Article 9 fund.

## 2. Assignment objective

The objective of this assignment is to design and demonstrate an **AI-assisted research process** that could support a sustainable investment team, asset manager, bank, consultancy, fintech or institutional investor.

You are not being asked simply to “use AI to pick green stocks”. You are being asked to design a **transparent, auditable and investment-relevant research pipeline** that combines:

- financial analysis;
- ESG data;
- climate and carbon metrics;
- biodiversity and nature-risk analysis;
- EU sustainable finance regulation;
- greenwashing checks;
- portfolio construction logic;
- human investment judgement.

The final output is a proposed sustainable portfolio and a professional explanation of **how you built it**.

## 3. Business problem

You are working as a sustainable investment research team. A European institutional client has asked you to build a sustainable equity portfolio. You are expected to design, document and defend an AI-assisted research process.

The client wants a portfolio that is financially credible, sustainability-aware and defensible under regulatory and reputational scrutiny.

Your task is to build a portfolio from the available company universe and explain the AI-agent research pipeline used to create it.

## 4. Core question

Your group must answer the following question:

**How can an AI-agent research pipeline be designed to build a sustainable investment portfolio that is financially credible, ESG-aware, climate-aware, biodiversity-aware and resistant to greenwashing?**

## 5. Group size

Groups should normally contain **4–5 students**.

Each group is responsible for organising its own workflow, assigning roles and ensuring that all members contribute meaningfully.

## 6. Investment mandate

Each group must define a clear investment mandate before building the portfolio.

Your mandate must specify:

| Item                     | Required explanation                                                                                                                                         |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Client                   | Who is the portfolio for? Pension fund, wealth client, foundation, university endowment, insurance company, Article 8-style fund, Article 9-style fund, etc. |
| Investment universe      | Which securities are eligible?                                                                                                                               |
| Portfolio type           | Long-only equity portfolio unless otherwise approved                                                                                                         |
| Number of holdings       | Recommended final portfolio: <b>15–25 securities</b>                                                                                                         |
| Time horizon             | Recommended: medium to long term                                                                                                                             |
| Benchmark                | Equal-weighted universe, market-cap-weighted universe, STOXX Europe 600, MSCI Europe, MSCI World, S&P 500, or another justified benchmark                    |
| Sustainability objective | ESG integration, net-zero transition, biodiversity-aware investing, EU Taxonomy alignment, impact orientation, controversy avoidance, or a combination       |
| Risk constraints         | Diversification, sector limits, maximum position size, carbon constraints, controversy exclusions                                                            |
| Final decision           | Which companies are included, excluded or placed on watchlist?                                                                                               |

## 7. Portfolio options

Each group should choose **one** of the following portfolio themes<sup>1</sup>.

### Option A — Sustainable equity portfolio

Build a diversified sustainable equity portfolio that improves ESG quality and manages climate, biodiversity and controversy risk.

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<sup>1</sup> **NB:** This is an academic prototype and must not be presented as a regulated investment product, financial advice or a compliant Article 8 / Article 9 fund.

## Option B — Net-zero transition portfolio

Build a portfolio that reduces carbon intensity while still financing credible transition companies. This option should avoid the simplistic approach of only buying already low-carbon companies.

## Option C — Biodiversity-aware portfolio

Build a portfolio that explicitly integrates nature-related dependencies, impacts, biodiversity risk, water risk, land-use risk and biodiversity-sensitive activities.

## Option D — ESG controversy-aware portfolio

Build a portfolio that combines financial analysis with AI-assisted controversy detection, greenwashing checks and reputational risk scoring.

## Option E — EU sustainable finance portfolio

Build a portfolio that explicitly considers EU Taxonomy eligibility, potential alignment, reported alignment where available, SFDR-style Principal Adverse Impact indicators, CSRD/ESRS-style disclosure and greenwashing risk.

## 8. Data

You will be provided with a structured course data pack. To build a complete multi-agent pipeline, you must supplement this with external financial data and unstructured public documents.

### 8.1 Minimum Data Requirements

Your research pipeline must ingest, process and document at least the following categories of data:

| <b>Data category</b>       | <b>Examples</b>                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------|
| Company identifiers        | Company name, ticker, ISIN, country, sector                                                    |
| Market data                | Prices, returns, volatility, drawdown, benchmark comparison                                    |
| ESG data                   | Environmental, social and governance indicators                                                |
| Climate data               | Scope 1, Scope 2, Scope 3 where available, carbon intensity, climate policy, emissions targets |
| Biodiversity / nature data | Biodiversity policy, water risk, land-use exposure, nature dependency, sector exposure         |

| Data category                    | Examples                                                                                                                                                                            |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EU Taxonomy / green revenue data | Taxonomy eligibility, potential alignment, limited reported alignment where available, green revenue proxy, DNSH/minimum-safeguards indicators, and sector mapping where available. |
| Controversy / greenwashing data  | News, regulatory issues, NGO concerns, company claims versus evidence                                                                                                               |

All datasets must be timestamped. Groups must state the data vintage used for ESG, governance, taxonomy, market prices and document extraction. If data vintages differ, the group must explain how this affects interpretation.

## 8.2 The Course Data Pack (Provided Structured Data)

You are provided with four core datasets containing cross-sectional ESG, regulatory, and mapping metrics. Your pipeline must ingest and utilize these files:

- i. equityBicsV2.csv (Identifiers & Sectors): Contains standard tickers, ISINs, and Bloomberg Industry Classification Systems (BICS) sector hierarchies. Use this to map identifiers and apply sector constraints to your portfolio.
- ii. esgEnvironmentalSocialConsolidatedV4.csv: Contains quantitative E and S metrics (e.g., GHG emissions, water usage, Scope 1-3).
- iii. esgGovernanceConsolidatedV4.csv: Contains corporate governance metrics (e.g., board diversity, executive compensation).
- iv. legalEntityEuTaxonomy.csv: **EU Taxonomy and green revenue overlay**: Contains EU Taxonomy eligibility metrics, potential alignment estimates, limited reported alignment metrics for revenue/capex/opex where available, and estimated DNSH/minimum-safeguards indicators.

Students must distinguish clearly between taxonomy eligibility, potential alignment and reported alignment. Because reported alignment coverage is sparse, this dataset should be used as a regulatory and green-revenue overlay rather than as a complete EU Taxonomy compliance dataset.

## 8.3 Financial & Market Data (To be sourced by students)

The provided course data pack **does not** contain historical financial or pricing data. To perform the financial analysis and portfolio optimisation requirements, your pipeline must obtain historical market data either through an approved open API or through a provided/cached market-data file. If APIs are used, the group must cache the data, document the download date, and explain any ticker-mapping issues.

- **Requirement:** You must use the identifier mappings in equityBicsV2.csv to fetch historical time-series pricing data via open APIs (e.g., Yahoo Finance via Yahoo Finance via `yfinance`, Alpha Vantage) to calculate returns, volatility, drawdowns, and covariance matrices.

## 8.4 Unstructured Document Corpus (To be curated by students)

To fulfill the "Controversy / greenwashing" and "Climate / Biodiversity policy" requirements, structured CSVs are insufficient. Your AI-agent pipeline will be evaluated on its ability to perform "document extraction," "controversy detection," and "greenwashing checks."

- **Requirement:** Your group must independently curate a local folder of unstructured documents for your AI agents to ingest, read, and cross-validate against the structured data.
- **Allowed Sources:** This folder should contain at least 15-20 recent, primary-source documents related to a subset of companies in your universe. You may use public external sources such as:
  - company annual reports;
  - company sustainability reports;
  - TCFD / climate reports;
  - CSRD / ESRS disclosures where available;
  - SBTi target information;
  - Transition Pathway Initiative data;
  - Climate Action 100+ benchmark data;
  - ENCORE nature-risk information;
  - WWF Biodiversity Risk Filter;
  - WRI Aqueduct water-risk data;
  - regulatory enforcement notices;
  - reputable news sources;
  - NGO or civil-society datasets.

The document corpus should cover at least 8–10 material companies from the candidate universe, including major final holdings, exclusions or watchlist names. It should include at least 10 primary company documents and may be supplemented by regulatory, NGO or reputable news sources.

## 8.5 Proprietary Data Rules

You are not required to use paid ESG datasets. If you use Bloomberg, Refinitiv, MSCI, Sustainalytics, Morningstar or any other licensed data source, you must confirm that you had legitimate access and explain exactly how the data were used.

## 9. AI-agent pipeline requirement

Your group must design an **AI-agent research pipeline**.

An “agent” does not need to mean a fully autonomous software agent. It may be:

- a Python module;
- a structured prompt workflow;
- an Excel + AI-assisted workflow;

- a notebook;
- a low-code workflow;
- an LLM-supported research process;
- a multi-agent framework;
- or a clearly documented sequence of AI-assisted research tasks.

The key requirement is that the pipeline must be **structured, repeatable, explainable and auditable**.

## 10. Minimum pipeline architecture

Your pipeline must include at least **eight** of the following agents or modules.

| Agent / module                             | Purpose                                                                                        |
|--------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>1. Mandate agent</b>                    | Defines the client, objective, benchmark, constraints and sustainability ambition              |
| <b>2. Data ingestion agent</b>             | Imports ESG, governance, price, taxonomy and external data                                     |
| <b>3. Data-quality agent</b>               | Flags missing values, stale data, outliers, inconsistent units and weak coverage               |
| <b>4. Document intelligence agent</b>      | Reads annual reports, sustainability reports or climate reports and extracts relevant evidence |
| <b>5. ESG scoring agent</b>                | Builds a transparent E, S, G and combined ESG score                                            |
| <b>6. Climate agent</b>                    | Calculates carbon intensity, WACI, emissions coverage and transition-readiness indicators      |
| <b>7. Biodiversity / nature-risk agent</b> | Assesses biodiversity exposure, nature dependency, water risk or sector-based nature impact    |
| <b>8. EU regulation agent</b>              | Reviews Taxonomy, SFDR, CSRD/ESRS or PAI relevance                                             |
| <b>9. Greenwashing agent</b>               | Flags vague claims, missing evidence, weak targets, controversies and inconsistencies          |
| <b>10. Financial analysis agent</b>        | Analyses returns, volatility, drawdown, Sharpe ratio, valuation or financial robustness        |
| <b>11. Portfolio construction agent</b>    | Selects securities and assigns weights using rules, ranking or optimisation                    |
| <b>12. Human review layer</b>              | Shows how the team challenged, corrected or overrode AI outputs                                |
| <b>13. Reporting agent</b>                 | Produces portfolio factsheet, investment memo, charts and presentation material                |

Your final report must include a diagram of your pipeline.

## 11. Portfolio construction requirements

Your final portfolio must meet the following minimum requirements unless your group justifies an alternative.

| Requirement                  | Minimum expectation                                                |
|------------------------------|--------------------------------------------------------------------|
| Candidate universe           | At least <b>50 companies</b> analysed                              |
| Final holdings               | <b>15–25 companies</b>                                             |
| Portfolio weights            | Must sum to 100%                                                   |
| Single-name limit            | No holding above 10% unless strongly justified                     |
| Sector diversification       | At least 5 sectors represented                                     |
| Benchmark comparison         | Required                                                           |
| Sustainability comparison    | Portfolio must be compared with the starting universe or benchmark |
| Carbon metric                | WACI or another carbon-intensity metric required                   |
| ESG metric                   | E, S, G and aggregate score required                               |
| Biodiversity / nature metric | At least one nature-risk proxy required                            |
| Greenwashing review          | Required                                                           |
| Exclusion / watchlist        | Required                                                           |
| Human judgement              | Required                                                           |

Your portfolio does not need to be the highest-returning portfolio, although desirable. It must be **defensible**.

## 12. Required portfolio outputs

Your final portfolio analysis must include:

1. final holdings and weights;
2. sector allocation;
3. country allocation if relevant;
4. top 5 holdings and rationale;
5. excluded companies and rationale;
6. ESG score versus benchmark or universe;
7. carbon intensity / WACI versus benchmark or universe;
8. biodiversity / nature-risk indicator;
9. controversy / greenwashing assessment;
10. return and risk comparison;
11. limitations and data-quality issues.

## 13. AI governance and responsible AI rules

You must disclose and govern your use of AI.

Your report must include an **AI Use Statement** covering:



| Area                  | Required disclosure                                                                    |
|-----------------------|----------------------------------------------------------------------------------------|
| AI tools used         | ChatGPT, Claude, Gemini, Copilot, Perplexity, Python libraries, agent frameworks, etc. |
| Agent roles           | What each AI agent/module was asked to do                                              |
| Prompts               | Include key prompts or workflow instructions in an appendix                            |
| Verification          | Explain how you checked AI outputs                                                     |
| Data provenance       | Identify where every important variable came from                                      |
| Human judgement       | Explain where humans accepted, rejected or modified AI outputs                         |
| Hallucination control | Explain how you avoided fake sources, fake numbers and unsupported claims              |
| Bias and limitations  | Discuss model bias, data bias, missing data and overfitting risks                      |

You must not:

- invent data;
- invent sources;
- cite papers or reports you have not checked;
- present AI-generated claims without verification;
- treat ESG ratings as objective truth;
- use paid or proprietary data without authorisation;
- present a black-box portfolio without explaining the methodology.

## 14. Written report / essay

### Length

**5,000–6,500 words**, excluding appendices, with a hard maximum of 8,000 words.

### Required title

**Methodology Report: AI-Agent Research Pipeline for Sustainable Portfolio Construction**

## Required structure

| Section                                       | Content                                                                           |
|-----------------------------------------------|-----------------------------------------------------------------------------------|
| <b>1. Executive summary</b>                   | State the client, mandate, final portfolio and key recommendation                 |
| <b>2. Investment mandate</b>                  | Define objective, benchmark, universe, sustainability ambition and constraints    |
| <b>3. Data sources and data quality</b>       | Explain datasets, missing data, assumptions, cleaning and limitations             |
| <b>4. AI-agent pipeline architecture</b>      | Explain each agent/module and how the workflow operates                           |
| <b>5. ESG methodology</b>                     | Explain E, S, G variables, scoring, normalisation and weights                     |
| <b>6. Climate methodology</b>                 | Explain carbon metrics, emissions coverage, transition indicators and limitations |
| <b>7. Biodiversity / nature methodology</b>   | Explain nature-risk proxy, biodiversity logic, dependencies and impacts           |
| <b>8. Greenwashing and controversy review</b> | Explain how claims were tested against evidence                                   |
| <b>9. Portfolio construction</b>              | Explain selection, weighting, constraints, optimisation or ranking process        |
| <b>10. Results</b>                            | Show final portfolio characteristics versus benchmark or universe                 |
| <b>11. Critical reflection</b>                | What worked, what failed, what would improve in a professional version            |
| <b>12. Conclusion</b>                         | Final investment committee recommendation                                         |

## Appendices

Your appendices should include:

- pipeline diagram;
- data dictionary;
- AI Use Statement;
- key prompts;
- code snippets or screenshots;
- portfolio holdings table;
- scorecard methodology;
- references.

## 15. Data dictionary requirement

Your report must include a data dictionary.

Example format:

| Variable            | Definition                              | Unit                            | Source                        | Extraction method            | Data type         | Confidence |
|---------------------|-----------------------------------------|---------------------------------|-------------------------------|------------------------------|-------------------|------------|
| Scope 1 emissions   | Direct GHG emissions                    | tCO <sub>2</sub> e              | Company sustainability report | AI extraction + manual check | Reported          | High       |
| Board independence  | Independent directors / total directors | %                               | ESG governance dataset        | Structured data              | Reported / vendor | Medium     |
| Biodiversity policy | Whether company has biodiversity policy | Binary / score                  | ESG dataset + report check    | Structured + AI review       | Reported / judged | Medium     |
| Carbon intensity    | Emissions / revenue                     | tCO <sub>2</sub> e / €m revenue | ESG data + financial data     | Calculation                  | Estimated         | Medium     |
| Greenwashing flag   | Claims unsupported by evidence          | 0/1 or low/medium/high          | Reports + news + human review | AI-assisted judgement        | Judgement-based   | Medium     |

Each important variable must be classified as one of the following:

- **reported;**
- **observed;**
- **estimated;**
- **AI-extracted;**
- **judgement-based.**

## 16. Presentation and demo

Each group will present during the final exam session.

### Timing

Recommended timing:

- **10 minutes presentation**
- **5 minutes Q&A**

The exact timing may be adjusted depending on the number of groups.

### Presentation structure

Your presentation should contain **8–10 slides**.

| Slide | Content                                   |
|-------|-------------------------------------------|
| 1     | Client mandate and investment objective   |
| 2     | AI-agent pipeline architecture            |
| 3     | Data sources and data-quality issues      |
| 4     | ESG, climate and biodiversity methodology |
| 5     | Greenwashing and controversy checks       |
| 6     | Portfolio construction process            |
| 7     | Final portfolio holdings and weights      |
| 8     | Portfolio versus benchmark                |
| 9     | Key limitations and human overrides       |
| 10    | Final investment committee recommendation |

## Demo requirement

You must show evidence of the pipeline working.

This may be:

- a live notebook;
- a short recorded demo;
- a workflow diagram with screenshots;
- an Excel model;
- a dashboard;
- a prompt-chain output;
- a code repository;
- or another transparent demonstration.

The demo does not need to be technically perfect. It must show that your group designed a real process rather than only describing one.

## 17. One-page portfolio factsheet

Each group must submit a **one-page portfolio factsheet**.

The factsheet should include:

- portfolio name;
- investment objective;
- benchmark;
- number of holdings;
- top 10 holdings;
- sector allocation;
- ESG score;
- carbon intensity / WACI;
- biodiversity / nature-risk indicator;
- controversy exclusions;
- return and volatility summary;

- investment rationale;
- key limitations.

## 18. Submission bundle

Each group must submit the following:

1. **Written report** in PDF format.
2. **Presentation deck** in PDF or PowerPoint format.
3. **One-page portfolio factsheet**.
4. **Technical appendix**, including data dictionary and AI Use Statement.
5. **Pipeline evidence**, such as code, notebook, Excel model, screenshots or recorded demo.
6. **References**.
7. **Individual contribution statement** from each student.

## 19. Individual contribution statement

Each student must submit a short statement of **150–250 words** explaining:

- their contribution to the group;
- which part of the pipeline they worked on;
- what they learned;
- whether AI was used in their contribution;
- any challenges faced.

This is not a separate essay, but it may be used to moderate marks where group contribution is uneven.

## 20. Required academic and professional references

Your report must cite at least eight serious sources, including at least three academic papers, two regulatory or practitioner sources, and the documentation for any major data source used.

At minimum, your references must include:

| Requirement                                       | Example                                                             |
|---------------------------------------------------|---------------------------------------------------------------------|
| One reading on ESG ratings or ESG data            | Aggregate Confusion                                                 |
| One reading on ESG alpha or sustainable investing | Honey, I Shrunk the ESG Alpha; Sustainable Investing in Equilibrium |
| One reading on climate finance or carbon risk     | Climate Finance; Do Investors Care About Carbon Risk?               |
| One reading on sustainable portfolio construction | ESG-efficient frontier; Roncalli climate portfolio construction     |
| One reading on biodiversity or nature finance     | Roncalli biodiversity notes; Dasgupta Review; TNFD                  |
| One regulatory or practitioner source             | EU Taxonomy, SFDR, CSRD/ESRS, TCFD, TNFD, SBTi, TPI                 |

You should use references to support your methodology, not just decorate the bibliography.

## 21. Assessment rubric

### Written report / essay — 60%

| Criterion                                     | Weight | What is being assessed                                                                          |
|-----------------------------------------------|--------|-------------------------------------------------------------------------------------------------|
| Investment mandate and problem framing        | 5%     | Clear client, objective, benchmark, constraints and sustainability ambition                     |
| Data quality and data provenance              | 10%    | Clear data sources, cleaning, missing data, assumptions and data dictionary                     |
| AI-agent pipeline design                      | 10%    | Logical, modular, transparent, auditable workflow                                               |
| ESG, climate and biodiversity methodology     | 15%    | Serious sustainability analysis, credible variables, clear scoring and awareness of limitations |
| Portfolio construction and financial analysis | 15%    | Defensible weights, risk/return analysis, benchmark comparison and diversification              |
| Critical reflection and limitations           | 5%     | Honest discussion of model risk, greenwashing, bias, missing data and professional improvements |

### Presentation, demo and Q&A — 40%

| Criterion                            | Weight | What is being assessed                                                         |
|--------------------------------------|--------|--------------------------------------------------------------------------------|
| Clarity of investment recommendation | 10%    | Clear final portfolio and investment committee message                         |
| Pipeline demonstration               | 10%    | Evidence that the AI-agent workflow exists and works at prototype level        |
| Professional presentation quality    | 10%    | Clear slides, good structure, strong storytelling and visual communication     |
| Q&A defence                          | 10%    | Ability to defend data, methodology, AI use, assumptions and portfolio choices |

## 22. Marking guidance

High-quality submissions will:

- define a precise investment mandate;
- avoid vague claims such as “this company is sustainable” without evidence;
- show a clear pipeline architecture;
- use AI responsibly and transparently;
- document data sources and weaknesses;
- distinguish ESG disclosure from ESG performance;
- integrate climate and biodiversity rather than treating them as decorative add-ons;
- compare the final portfolio with a benchmark or baseline;
- explain exclusions and human overrides;
- discuss greenwashing and model risk seriously;
- produce a professional investment committee-style recommendation.

Weak submissions will:

- rely only on generic AI output;
- use unsupported claims;
- provide no data dictionary;
- ignore missing data;
- treat one ESG score as objective truth;
- build a portfolio without explaining weights;
- ignore financial risk;
- ignore biodiversity or nature risk;
- include fake or unchecked citations;
- present a black-box model;
- fail to explain how AI was governed.

## 23. Suggested project timeline

| Date                                   | Milestone                                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Friday 24 April 2026</b>            | Groups formed. Assignment launched. Select portfolio theme and provisional mandate.                                 |
| <b>Friday 8 May 2026</b>               | Submit one-page proposal: mandate, universe, data sources, proposed AI-agent architecture.                          |
| <b>Friday 15 May 2026</b>              | Prototype clinic: show early data, draft scoring model and pipeline design.                                         |
| <b>Friday 22 May 2026, afternoon</b>   | Presentation rehearsal and investment committee challenge.                                                          |
| <b>Friday 22 May 2026, 18:15–20:15</b> | Final presentation, demo and Q&A.                                                                                   |
| <b>Final deadline</b>                  | Written report, factsheet and technical appendix due by the final presentation session unless otherwise instructed. |

## 24. Professional relevance

This assignment is designed to help you build a project that is useful beyond the course.

A strong student should be able to describe the project on a CV or in an interview as follows:

**Built an AI-agent research pipeline for sustainable portfolio construction, integrating ESG data, carbon metrics, biodiversity risk, greenwashing checks, financial analysis and portfolio optimisation. Produced an investment committee-style recommendation, methodology report and live group presentation.**

A more technical version would be:

**Designed a multi-agent AI workflow for ESG and climate investment research, including data ingestion, document extraction, sustainability scoring, controversy detection, portfolio constraints and human-in-the-loop validation.**

The purpose of this project is not to claim that AI can replace investment judgement. The purpose is to show that you can combine **finance, sustainability, data, AI and critical judgement** into a defensible research process.

## 25. Final reminder

Your final portfolio is important, but the real assessment is your **process**.

A good submission should be able to answer:

“Would a serious investment committee, asset manager, regulator or client understand how this portfolio was built, trust the evidence, and see clearly where human judgement entered the process?”

That is the standard you should aim for.